Huawei Access Node QoS Congestion Event (WFD-ACC-09) SOP

Node: WFD-ACC-09 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_2456

Description

The QoS congestion event on the Huawei Access Node WFD-ACC-09 at the Whitefield POP can be attributed to microbursts and interface tail drops caused by the oversubscription of the access interface, leading to QoS queue starvation. This phenomenon occurs due to the packet buffering mechanism on the Huawei Access Node, which relies on the egress queue management to regulate the outgoing traffic. When the access interface is oversubscribed, the egress queues become congested, resulting in the tail drop of packets and the subsequent QoS queue starvation. This can lead to a critical congestion event, triggering the ALARM_2456 alarm code. To mitigate this issue, it is essential to monitor the access interface utilization and ensure that the egress queue management is properly configured to prevent oversubscription.

Pre-requisite Checks

- Check: Verify the redundant link status on the Huawei Access Node WFD-ACC-09 to confirm that it is operating in standby mode.

- Check: Confirm that the access interface is not oversubscribed, and the egress queue management is properly configured.

- Check: Ensure that the Huawei Access Node WFD-ACC-09 is running the latest software version.

Potential Root Causes

Cause	Probability
Egress queue starvation due to oversubscription of the access interface.	High
Inadequate egress queue management configuration.	Medium
Packet buffering mechanism malfunction on the Huawei Access Node.	Low

Diagnostic Commands & Expected Outputs

```
Command: display interface GigabitEthernet 0/0/1, Expected Output: Interface GigabitEthernet 0/0/1 is up, line protocol is up, Description: This command is used to verify the operational status of the access interface GigabitEthernet 0/0/1. The expected output indicates that the interface is up and running, but it does not provide any information about the egress queue management configuration.

Command: display qos queue GigabitEthernet 0/0/1, Expected Output: Queue GigabitEthernet 0/0/1: 10 packets, 1000 bytes, Description: This command is used to verify the egress queue configuration on the access interface GigabitEthernet 0/0/1. The expected output indicates that
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```
the queue is congested, with 10 packets and 1000 bytes of data.

Command: display qos queue GigabitEthernet 0/0/1 detail, Expected
Output: Queue GigabitEthernet 0/0/1: 10 packets, 1000 bytes, 0 drops,
Description: This command is used to verify the egress queue
configuration on the access interface GigabitEthernet 0/0/1 in detail.
The expected output indicates that the queue is congested, with 10
packets and 1000 bytes of data, and 0 drops.
```

Resolution Steps

Step 1: Step: Configure egress queue management, Command: queue-gigabitethernet 0/0/1 egress-queue min-threshold 100 max-threshold 200, Description: This command is used to configure the egress queue management on the access interface GigabitEthernet 0/0/1. The min-threshold and max-threshold values are set to 100 and 200, respectively, to prevent oversubscription and egress queue starvation.

Step 2: Step: Verify egress queue configuration, Command: display qos queue GigabitEthernet 0/0/1 detail, Description: This command is used to verify the egress queue configuration on the access interface GigabitEthernet 0/0/1 in detail. The expected output indicates that the queue is properly configured, with the min-threshold and max-threshold values set to 100 and 200, respectively.

Post-Check Verification

```
Command: display interface GigabitEthernet 0/0/1, Expected Output:
Interface GigabitEthernet 0/0/1 is up, line protocol is up,
Description: This command is used to verify the operational status of
the access interface GigabitEthernet 0/0/1. The expected output
indicates that the interface is up and running, and the egress queue
management is properly configured.

Command: display qos queue GigabitEthernet 0/0/1, Expected Output:
Queue GigabitEthernet 0/0/1: 0 packets, 0 bytes, Description: This
command is used to verify the egress queue configuration on the access
interface GigabitEthernet 0/0/1. The expected output indicates that
the queue is empty, and the egress queue management is properly
configured.
```

Roll-back Procedures

- Step: Restore original egress queue configuration, Command: undo queue-gigabitethernet 0/0/1 egress-queue min-threshold 100 max-threshold 200, Description: This command is used to restore the original egress queue configuration on the access interface GigabitEthernet 0/0/1. The min-threshold and max-threshold values are restored to their original settings.